

#1. Write a program to print the first N natural numbers. Input : 10 , Output : 1 2 3 4 5 6

```
7 8 9 10
'''
n=int(input("Enter the number"))
for i in range(1, n+1):
    print(i, end=" ")
i=1
n=int(input("Enter the Number: "))
while i<=n:
    print(i,end=" ")
    i+=1
'''
```

#2. Write a program to print the first N even natural numbers. Input : 5 , Output 2 4 6 8 10

```
'''
n=int(input("Enter the Number:"))
for i in range(1,n+1):
    print(i*2, end=" ")
i=1
n=int(input("Enter the Number:"))
while i<=n:
    print(i*2, end=" ")
    i+=1
'''
```

#3. Write a program to print the first N odd natural numbers. Input : 5 , Output : 1 3 5 7 9

```
'''
n=int(input("Enter the Number in for loop:"))
for i in range(1,n+1):
    print(i*2-1, end=" ")
print("\n")
i=1
n=int(input("Enter the Number in (while loop):"))
while i<=n:
    print(i*2-1, end=" ")
    i+=1
'''
```

#4. Write a program to print first N multiples of 3. Input : 7 , Output : 3 6 9 12 15 18 21

```
'''
n=int(input("Enter the Number:"))
for i in range(1,n+1):
    print(i*3, end=" ")
print("\n")
i=1
n=int(input("Enter the Number:"))
while i<=n:
    print(i*3, end=" ")
    i+=1
'''
```

#5. Write a program to print first N multiples of 5. Input : 5 , Output : 5 10 15 20 25

```
'''
n=int(input("Enter the Number:"))
for i in range(1,n+1):
    print(i*5, end=" ")
print("\n")
i=1
n=int(input("Enter the Number:"))
while i<=n:
    print(i*5, end=" ")
    i+=1
'''
```

#6. Write a program to print all the multiples of 2 till N. Input : 15 , Output : 2 4 6 8 10 12 14

```
'''
n=int(input("Enter the Number:"))
for i in range(1,n+1):
    if i%2==0:
        print(i)
print("\n")
i=1
n=int(input("Enter the Number:"))
while i<=n:
    if i%2==0:
        print(i)
    i+=1
'''
```

#7. Write a program to print all the numbers which are multiples of either 2 or 3 till N. Input : 15 , Output : 2 3 4 6 8 9 10 12 14 15

```
'''
n=int(input("Enter the Number:"))
for i in range(1,n+1):
    if i%2==0 or i%3==0:
        print(i)
```

```

print("\n")
i=1
n=int(input("Enter the Number:"))
while i<=n:
    if i%2==0 or i%3==0:
        print(i)
    i+=1
'''

#8. Write a program to print all the numbers which are multiples of either 2, 5 or 7 till N.
#Input : 15 , Output : 2 4 5 6 7 8 10 12 14 15
'''

n=int(input("Enter the Number:"))
for i in range(1,n+1):
    if i%2==0 or i%5==0 or i%7==0:
        print(i)

i=1
n=int(input("Enter the number: "))
while i<=n:
    if i%2==0 or i%5==0 or i%7==0:
        print(i)
    i+=1
'''

#9. Write a program to print the first N multiples of either 3, 5 or 7. Input : 10 , Output : 3
#5 6 7 9 10 12 14 15 18
'''

n=int(input("Enter the Number:"))
for i in range(1,n+1):
    if i%3==0 or i%5==0 or i%7==0:
        print(i)

i=1
n=int(input("Enter the number: "))
while i<=n:
    if i%3==0 or i%5==0 or i%7==0:
        print(i)
    i+=1
'''

#10. Find the sum of all digits in a positive integer. Input : 123456789 , Output : 45
'''

num=int(input("Enter the number;"))
s=0
for i in str(num):
    s+=int(i)
print("Sum of given num for loop:",s)
print()
num=int(input("Enter the Number:"))
s=0
while num>0:
    s+= num % 10
    # print(" num of %:",num)
    num //= 10
    # print("num of floor DIV:",num)
    print("sum of given numbers:",s)
'''

#11. Count the number of digits in a positive integer. Input : 123456789 , Output : 9
'''

num=int(input("enter the Number;"))
count = 0
while num > 0:
    count += 1
    num //= 10
print("COUNT OF :",count)
print("\n")
num=int(input("Enter the number;"))
count=0
for i in str(num):
    count+=1
print("COUNT OF :",count)
'''

#12. Write a program to find factors of a given number. Input : 20 , Output : 1 2 4 5 10 20
'''

num=int(input("Enter the Numbmer:"))
print("Factors of",num,"using for loop:")
for i in range(1,num+1):
    if num%i==0:
        print(i,end=" ")
print()
num=int(input("Enter the Numbmer:"))
i=1
print("Factors of", num, "using while loop:")

```

```

while i<=num:
    if num%i==0:
        print(i,end=" ")
    i += 1
'''

#13. Write a program to count factors of a given number. Input : 20 , Output : 6
num=int(input("Enter the number;"))
count=0
for i in range(1,num+1):
    if num%i==0:
        count+=1
print("number of factors(For LOOP):",count)
print()
num=int(input("Enter the Number:"))
i=1
count=0
while i<=num:
    if num%i==0:
        count+=1
    i+=1
print("number of factors(While LOOP):",count)
'''

#14. Write a program to find whether the given number is a prime number or not.I nput :
#11 , Output : Yes
num=int(input("Enter the number;"))
c=0
for i in range(1,num+1):
    if num%i==0:
        c+=1
if c==2:
    print("prime number")
else:
    print("not prime number")

'''

#15.Print all prime numbers from 2 up to and including 'n'. Input : 15 , Output : 2 3 5 7 11 13
'''
num=int(input("Enter the number;"))
for i in range(2,num+1):
    c=0
    for j in range(1,i+1):
        if i%j==0:
            c+=1
    if c==2:
        print("yes prime number",i)
'''

#16. Write a program to find the greatest common factor of given 2 integers. Input : 10 20 ,Output : 10
'''
a=int(input("Enter the number;"))
b=int(input("Enter the number:"))
gcf=1
for i in range(1,min(a,b)+1):
    if a%i==0 and b%i==0:
        gcf=i
print("GCF of", a, "and", b, "is:", gcf)
'''

#17.Print the common factors of two positive integers n and m. Input : 8 12 , Output : 1 2 4
'''
a=int(input("Enter the number;"))
b=int(input("Enter the number:"))
for i in range(1,min(a,b)+1):
    if a%i==0 and b%i==0:
        print(i,end="")
'''

#18. Write a program to get the Fibonacci series between 0 to 50.
'''
a=int(input("Enter the number:"))
n1=0
n2=1
for i in range(1,a+1):
    print(n1)
    n3=n1+n2
    n1=n2
    n2=n3
'''

#19. Write a program to find the factorial of a given number.
'''
n=int(input("Enter the number:"))
fact=1

```

```

for i in range(1,n+1):
    fact*=i
print("Factorial of", n, "is:", fact)
print()
n=int(input("enter the Number:"))
while i<n:
    fact*=i
print("Factorial of", n, "is:", fact)
'''

#20. Write a program that reads a set of integers, and then prints the sum of the even and odd integers.
'''

num=int(input("Enter the number;"))
e=0
o=0
for i in range(num):
    if i%2==0:
        e+=i
    else:
        o+=i
print("Sum of given odd num :",o)
print("Sum of given even num :",e)
'''

#21. Write a program to check whether the given number is palindrome or not.
##Palindrome is a number when reversed it gives the same number. Example 1: If the
##number is 121 Reverse is 121 Hence, 121 is a Palindrome Example 2: If the number is
##438 Reverse is 834 Hence, 438 is not a Palindrome
'''

num=int(input("Enter the number:"))
s=0
a=num
t=a
while a>0:
    r=a%10
    s=(s*10)+r
    a//=10
if s==t:
    print("YES")
else:
    print("NO")
'''

#22. Write a program to print if a number is an Armstrong number or not. 153 is an
##Armstrong number. 13+53+33 = 153
'''

a=int(input("Enter the Number:"))
c=0
temp=a
while a>0:
    a//=10
    c+=1
a=temp
arm=0
while a>0:
    r=a%10
    arm+=r**c
    print(arm)
    a//=10
if temp==arm:
    print("YES ,is arm number:")
else:
    print("NO, is not arm number:")
'''

#23 patterns
#23(1)
'''

for i in range(1,6):
    for j in range(1,6):
        if i==1 or i==5 or j==1 or j==5:
            print("*",end=" ")
        else:
            print(" ",end=" ")
    print()
'''

#23(2)
'''

for i in range(1,6):
    for j in range(1,6):
        print("*",end=" ")
    print()
'''

```

```
#23(3)
'''
for i in range(1,6):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
'''

#23(4)
'''
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
'''

#23(5)
'''
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
'''

#23(6)
'''
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(i,end=" ")
    print()
'''

#23(7)
'''
n=6
for i in range(1,n+1):
    for j in range(1,n-i+1):
        print("*",end=" ")
    print()
'''

#23(8)
'''
n=5
for i in range(1,n+1):
    for j in range(1,n-i+1):
        print(" ",end=" ")
    for k in range(1,i+1):
        print("*",end=" ")
    print()
'''

#23(9)
'''
n=6
for i in range(1,n+1):
    for j in range(1,i+1):
        print(" ",end=" ")
    for k in range(1,n-i+1):
        print("*",end=" ")
    print()
'''

#23(10)
'''
for i in range(1,6):
    for j in range(1,6-i+1):
        print("*",end=" ")
    print()
for i in range(1,6):
    for j in range(1,i+1):
        print("*", end=" ")
    print()
'''

#23(11)
'''
for i in range(1,6):
    for j in range(1,6-i+1):
        print(" ",end=" ")
    for k in range(1,6):
        print("*",end=" ")
    print()
'''
```

```

#23(12)
'''
n=5
for i in range(1,n+1):
    for j in range(1,n-i+1):
        print(" ",end=" ")
    for k in range(0,2*i-1):
        print("*",end=" ")
    print()
for i in range(n-1,0,-1):
    for j in range(0,n-i):
        print(" ",end=" ")
    for k in range(0,2*i-1):
        print("*",end=" ")
    print()
'''

```

```

#23(13)butterfly
'''
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print("*",end=" ")
    p=n-i+1
    for k in range(0,2*p):
        print(" ",end=" ")
    for l in range(1,i+1):
        print("*",end=" ")
    print()
for i in range(1,n-1):
    for j in range(0,2*n+2):
        print("*",end=" ")
    print()
for i in range(n,0,-1):
    for j in range(1,i+1):
        print("*",end=" ")
    p=n-i+1
    for k in range(1,2*p+1):
        print(" ",end=" ")
    for l in range(1,i+1):
        print("*",end=" ")
    print()
'''

```

```

#23(14)
'''
n=5
for i in range(1,n+1):
    for j in range(1,n-i+1):
        print(" ",end=" ")
    for k in range(0,2*i-1):
        print("*",end=" ")
    print()
'''

```

```

#23(15)Diamond shape
'''
n=5
for i in range(1,n+1):
    for j in range(1,n-i+1):
        print(" ",end=" ")
    for k in range(1,2*i):
        print("*",end=" ")
    print()
for i in range(n-1,0,-1):
    for j in range(1,n-i+1):
        print(" ",end=" ")
    for k in range(1,2*i):
        print("*",end=" ")
    print()
'''

```

```

#23(16)
'''
n=6
for i in range(1,n+1):
    for j in range(1,i+1):
        print(" ",end=" ")
    for k in range(1,n-i+1):
        print(j,end=" ")
    print()
n=6

```



```
for i in range(1,n+1):
    for j in range(1,i+1):
        print(" ",end=" ")
    for k in range(1,n-i+1):
        print(i,end=" ")
    print()
'''
```